Advanced Astronomy: Shared Standards (AA:X)

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Advanced Astronomy

Advanced Astronomy (AA) is structured as two semester-long elective classes covering astronomical phenomena and the science behind their observation and explanation. There are three elements to the course. The first two are the semester electives

- AA: Solar System (AA:SS)
  - Fall semester elective
- AA: Universe (AA:U)
  - Spring semester elective

The third concerns the astronomical and physical phenomena common to both classes. For example, the properties and workings of the Sun are covered in AA:SS, but solar-mass stars make up an important starting point for the analysis of stars in general covered in AA:U. For this reason there is another component common to both

- AA: X
    - Fall or Spring term; whichever comes first

This material is taken once, the first time it is encountered. For example, if a student takes only AA:U or takes it before AA:SS, then they will cover the material common to both the Sun and solar-mass stars in AA:U.

General

Students will practice physical and mathematical concepts in the context of astronomy. Formative work will include

- Textbook and other assigned reading
- Proper note taking during reading and lessons
- Construction of simple models for
    - demonstrating concepts
    - representation of data
    - astronomical measurement
- Participation in mock quizzes
- Completion of Jupyter Notebook packets
    - Basic LaTeX typesetting
    - Astronomical computation with Python

Summative work will include

- Tests and quizzes
- Graded Jupyter Notebooks
- Other appropriate demonstrations of knowledge such as
    - Presentations of findings
    - Coherent formulation of verbal arguments

Thus, students will demonstrate knowledge and understanding of the material in their class. A specific list of topics from which the material will be pulled follows.

AA: X

This document collects the shared standards used across Advanced Astronomy. These objectives may be taught in either semester, whichever comes first, and then treated as prerequisite background thereafter.

Note on VLO codes

Each code has the form `aa.<course>.<unit>.<topic>.<objective>`, where `<course>` is `x`, `u`, or `ss`. These codes are meant to identify major assessable objectives and objective-clusters. They do **not** attempt to enumerate every vocabulary item, bolded term, or small subskill that students are still responsible for learning from packets, notes, lessons, and examples.

Euclidean Geometry

Topic	I am able to…	Code (`aa.x.euclidean_geometry.`)
Triangles	… use sines, cosines, tangents, and Pythagoras' Theorem to solve right triangles quantitatively.	`triangles.010`
	… identify which geometric tool is appropriate in a given applied problem.	`triangles.020`
	… apply right-triangle geometry to real astronomical estimation problems involving lengths, distances, and angles.	`triangles.030`
Radians	… explain what a radian measures geometrically.	`radians.010`
	… use radians to determine arc lengths and related quantities.	`radians.020`
Units	… convert between degrees and radians.	`units.010`
	… define parsecs (pc) and astronomical units (au), and convert among those units and meters.	`units.020`

Spectroscopy

Topic	I am able to…	Code (`aa.x.spectroscopy.`)
Electricity	… explain and model the phenomena described by Coulomb's Law qualitatively, including inverse-square behavior and comparison to gravitation.	`electricity.010`
	… calculate electric forces and accelerations in simple Coulomb-law problems.	`electricity.020`
	… describe and use electric field lines qualitatively.	`electricity.030`
	… define the fine-structure constant as a dimensionless quantity built from fundamental constants.	`electricity.040`
Photoelectric effect	… explain the photoelectric effect and why it implies that light cannot be understood purely as a classical wave.	`photoelectric_effect.010`
	… state and use Einstein's relation for photon energy in terms of frequency.	`photoelectric_effect.020`
Hydrogen atom	… write de Broglie's relation, explain what it means, and use it quantitatively.	`hydrogen_atom.010`
	… explain Bohr's picture of the hydrogen atom in light of de Broglie and Einstein.	`hydrogen_atom.020`
	… identify and use the ground-state energy of hydrogen and explain why the spectrum is discrete.	`hydrogen_atom.030`
	… express major hydrogen-atom quantities such as the ground-state energy, Bohr radius, and electron speed in terms of the fine-structure constant.	`hydrogen_atom.040`

Topic	I am able to…	Code (`aa.x.spectroscopy.`)
	… identify major hydrogen transitions and explain what is special about the familiar hydrogen line series.	`hydrogen_atom.050`
Quantum mechanics	… explain the wave function as a probability-amplitude object and describe the role of the Schrödinger equation and stationary states in the modern picture of the atom.	`quantum_mechanics.010`
	… connect the modern quantum picture to the semi-classical one by relating de Broglie ideas, momentum and wave number, and transitions between atomic energy levels.	`quantum_mechanics.020`
Spectroscopy	… explain the Rydberg formula and use it to calculate wavelengths associated with hydrogen transitions.	`atomic_spectroscopy.010`
	… explain emission and absorption spectra, and explain Kirchhoff's laws in terms of transitions between atomic energy levels.	`atomic_spectroscopy.020`
	… explain what the Bohr magneton is and use it in simple calculations.	`atomic_spectroscopy.030`
	… explain degeneracy and how it is lifted by a magnetic field in the Zeeman effect.	`atomic_spectroscopy.040`
	… describe line broadening, identify major broadening mechanisms, and explain what they reveal about temperature, pressure, and magnetic field.	`atomic_spectroscopy.050`

Topic	I am able to…	Code (`aa.x.spectroscopy.`)
	… explain why molecular rotational and vibrational modes lead to additional spectral lines and extend this logic to new spectra when an allowed energy pattern is given.	`atomic_spectroscopy.060`

Waves and Light

Topic	I am able to…	Code (`aa.x.waves_light.`)
Simple harmonic oscillation	… explain simple harmonic motion, period, frequency, wavelength, and their angular versions.	`sho.010`
	… connect simple harmonic oscillation to plane-wave propagation.	`sho.020`
	… explain wave propagation in terms of transverse vs. longitudinal motion, dispersion relation, and wave speed.	`sho.030`
	… describe a plane wave quantitatively in terms of amplitude, angular frequency, wave number, and phase.	`sho.040`
Spectrum and spectral classification	… distinguish the major regions of the electromagnetic spectrum and state the approximate wavelength range visible to the human eye.	`spectrum.010`
Doppler	… describe the Doppler shift quantitatively and explain its origin.	`doppler.010`

Topic	I am able to…	Code (aa.x.waves_light.)
	… interpret redshift and blueshift quantitatively and apply Doppler shifts to spectroscopic lines in order to determine radial motion.	doppler.020
Optics	… describe reflection, transmission, refraction, diffraction, interference, and superposition at a conceptual level.	optics.010
	… use Snell's law of refraction.	optics.020
	… explain Huygens' principle and its role in diffraction.	optics.030
	… explain resolution, diffraction limitation, the Airy disk, and Rayleigh's criterion.	optics.040
	… describe the role of aperture and the primary objective in telescopy and use diffraction arguments to estimate the aperture required to resolve an astronomical target.	optics.050

Dimensional Analysis

Topic	I am able to…	Code (aa.x.dimensional_analysis.)
(In)significant figures and uncertainties	… understand the size of uncertainties and represent physical values accordingly.	uncertainties.010
	… round to a reasonable number of figures when the uncertainty is not stated or is unknown.	uncertainties.020

Topic	I am able to…	Code (`aa.x.dimensional_analysis.`)
Dimensions vs. units	… distinguish dimensions such as length, mass, and time from units such as meters, kilograms, and seconds.	`units.010`
	… convert between common units and choose reasonable units for very large or very small quantities.	`units.020`
SI, dex, metric prefixes	… convert to and from SI units and interpret decimal exponents.	`prefixes.010`
	… interpret common metric prefixes across a wide range of scales.	`prefixes.020`
Physical quantities	… explain the meaning of fundamental dynamical quantities such as mass, distance, time, velocity, momentum, acceleration, force, energy, and power.	`quantities.010`
	… explain the distinction between mass and weight.	`quantities.020`
	… determine dimensions and SI units of major dynamical quantities and convert between SI and common astronomical units.	`quantities.030`
Fundamental constants	… explain the meaning of the constants G, h, and c and determine their dimensions.	`constants.010`
	… use dimensional analysis to check or guess formulae involving fundamental constants.	`constants.020`
Applied dimensional analysis	… explain proportionality and constants of proportionality.	`applied.010`
	… use dimensional analysis to check familiar formulae and to guess plausible formulae in unfamiliar situations.	`applied.020`

Topic	I am able to…	Code (aa.x.dimensional_analysis.)
	… use dimensional analysis to build physical intuition about the structure and timescales of familiar and unfamiliar systems.	applied.030

Astrodynamics

Topic	I am able to…	Code (aa.x.astrodynamics.)
Newton's First Law	… explain what an inertial frame is and why it matters.	n1.010
	… use conservation of linear momentum in simple before-and-after situations.	n1.020
Newton's Second Law	… recognize that Newton's Second Law holds only in inertial frames.	n2.010
	… recognize and use Newton's Second Law in multiple equivalent forms.	n2.020
Newton's Third Law	… explain Newton's Third Law and use it in simple problems.	n3.010
Newton's Law of Universal Gravitation	… identify Newton's law of gravitation, explain its functional form, and determine the dimensions of Newton's constant.	gravity.010
Uniform circular motion	… explain the directions and magnitudes of velocity and acceleration in uniform circular motion.	ucm.010
	… solve for speed, period, or radius given the other two.	ucm.020
	… state and use the centripetal acceleration in circular-orbit problems.	ucm.030

Topic	I am able to…	Code (`aa.x.astrodynamics.`)
Energy	… state the dimensions of energy and identify common forms such as kinetic and gravitational potential energy.	`energy.010`
	… explain that energy is conserved and use energy conservation in simple problems.	`energy.020`
	… state the Virial Theorem and apply it in simple settings.	`energy.030`
	… explain how the two-body problem reduces to an effective one-body problem about the center of mass.	`energy.040`
Combinations	… combine Newton's laws to derive gravitational field strength, orbital speed, Newton's version of Kepler's Third Law, and escape speed.	`combinations.010`
	… solve the resulting relations for any of the main dynamical variables.	`combinations.020`
	… apply these relations numerically to determine mass, distance, speed, acceleration, gravitational field, orbital speed, or escape speed.	`combinations.030`
	… state what geosynchronous and geostationary orbits are.	`combinations.040`
Center of mass	… produce and use a formula for the center of mass of two massive bodies.	`center_of_mass.010`
	… calculate the center of mass in appropriate units and express its location relative to the radii of the bodies involved.	`center_of_mass.020`